

Electricity and Circuits

Subject: Physics

Grade: Grade 9

Understand current, voltage, resistance, and how simple circuits are built and measured.

Current, Voltage, and Resistance

Electric current is the flow of charge. Voltage pushes charges through the circuit, and resistance slows the flow.

Simple Circuits

A basic circuit has a power source, conductor, and load. Current flows only when the circuit is closed.

Series and Parallel Circuits

In series circuits, current follows one path. In parallel circuits, it can split into multiple branches.

Safety and Everyday Use

Electricity must be used safely. Switches, fuses, and insulating materials protect people and devices.